BRIDGES IN MATHEMATICS • KINDERGARTEN • UNIT 4

Paths to 20: Comparing Groups on the Number Rack

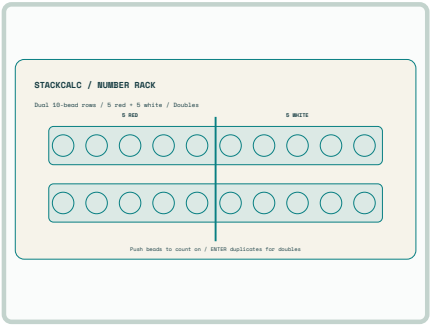
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.CC.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Push 8 beads on top wire, 5 beads on bottom wire. Compare lengths visually.

Sketch Model on Paper:
Draw lines connecting top and bottom beads to see the difference.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	9	X: 9	Key top wire count i
2.	ENTER	Y: 9 | X: 9	Stack lift copies 9
3.	6	Y: 9 | X: 6	Key bottom wire coun
4.	-	X: 3	Difference: 9 - 6 =

EXPECTED LCD X: 3 more beads

1 **Top row has 9, bottom row has 5. Difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Top row has 8, bottom row has 6. Difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Top row has 10, bottom row has 5. Difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Both rows have 8 beads. Difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 4

Paths to 20: Comparing Groups on the Number Rack • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Number Rack (20-Bead Dual Rod)

CAD Blueprint: number_rack

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White Frame with Signal Red & White Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Top row has 9, bottom row has 5. Difference?** Standard: CCSS.MATH.K.CC.C.6	4	[9] [ENTER] [5] [-]
2	**Top row has 8, bottom row has 6. Difference?** Standard: CCSS.MATH.K.CC.C.6	2	[8] [ENTER] [6] [-]
3	**Top row has 10, bottom row has 5. Difference** Standard: CCSS.MATH.K.CC.C.6	5	[1][0] [ENTER] [5] [-]
4	**Both rows have 8 beads. Difference?** Standard: CCSS.MATH.K.CC.C.6	0	[8] [ENTER] [8] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Reinforce that subtraction measures the space between two quantities.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.